EXPERIMENT NO. 7

AIM: To study and verify the truth table of various logic gates.

APPARATUS REQUIRED: Logic gates (IC) trainer kit, Connecting patch chords.
IC 7400, IC 7408, IC 7432, IC 7406, IC 7402, IC 7404, IC 7486

THEORY:

The basic logic gates are the building blocks of more complex logic circuits. These logic gates perform the basic Boolean functions, such as AND, OR, NAND, NOR, Inversion Exclusiv-OR, Exclusive-NOR. Fig. below shows the circuit symbol, Boolean function, and truth. It is seen from the Fig that each gate has one or two binary inputs, A and B, and one binary output, C. The small circle onthe output of the circuit symbols designates the logic complement. The AND, OR, NAND, and NOR gates can be extended to have more than two inputs. A gate can be extended to have multiple inputs if the binary operation it represents is commutative and associative.

These basic logic gates are implemented as small-scale integrated circuits (SSICs) or as part of more complex medium scale (MSI) or very large-scale (VLSI) integrated circuits. Digital IC gates are classified not only by their logic operation, but also the specific logic-circuit family to which they belong. Each logic family has its own basic electronic circuit upon which more complex digital circuits and functions are developed. The following logic families are the most frequently used.

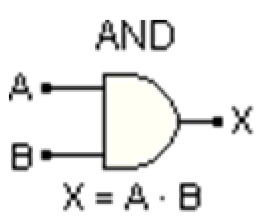
PROCEDURE:

1. Check the components for their working.
2. Insert the appropriate IC into the IC base.
3. Make connections as shown in the circuit diagram.
4. Provide the input data via the input switches and observe the output on output LEDs.

AND GATE:-

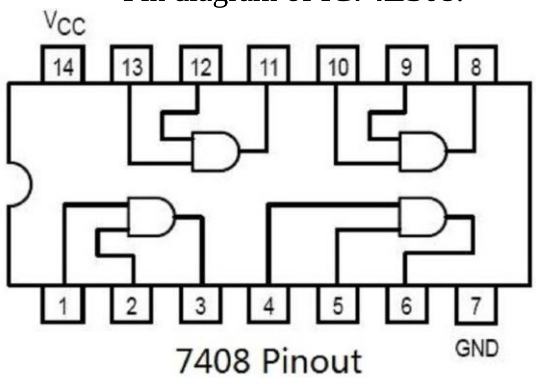
An AND gate has two or more inputs but only one output. The output assumes the logic 1 state only when each one of its inputs is at logic 1 state. The output assumes logic 0 state even if one of its input is at logic 0 state. AND gate is also called an „all or nothing‟ gate.

The logic symbol & truth table of two input AND gate are shown in figure



A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

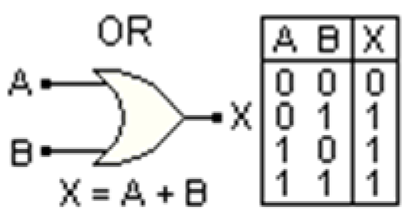
Pin diagram of IC**74LS08**:



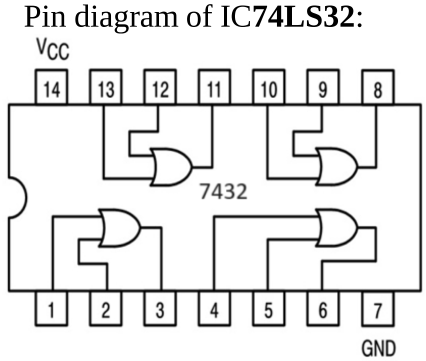
OR GATE

Like an AND gate, an OR gate may have two or more inputs but only one output. The output assumes the logic 1 state, even if one of its inputs is in logic 1 state. Its output assumes logic 0 state, only when each one of its inputs is in logic 0 state. OR gate is also called an „any or all‟ gate. It can also be called an inclusive OR gate because it includes the condition „both the input can be present‟.

The logic symbol & truth table of two input OR gate are shown in figure.



Pin diagram of IC**74LS32**:



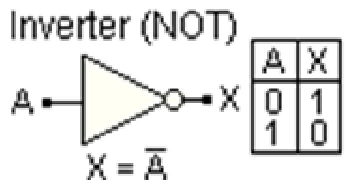
NOT GATE

A NOT gate is also known an inverter, has only one input and only one output. It is a device whose output is always the complement of its input. That is the output of a not gate assumes the logic 1 state when its input is in logic 0 state and assumes the logic 0 state when itsinput is in logic 1 state.

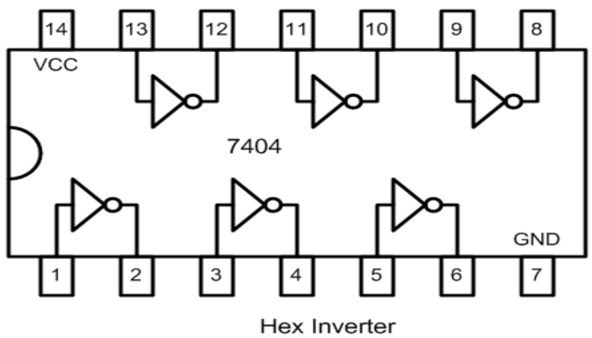
The logic symbol & truth table of NOT gate are shown in figure.
With input variable A the Boolean expression for output can be written as;

$$X = \overline{A}$$

This is read as "X is equal to A bar".



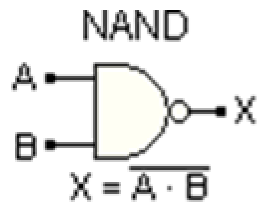
Pin diagram for IC**74LS04**:



NAND GATE

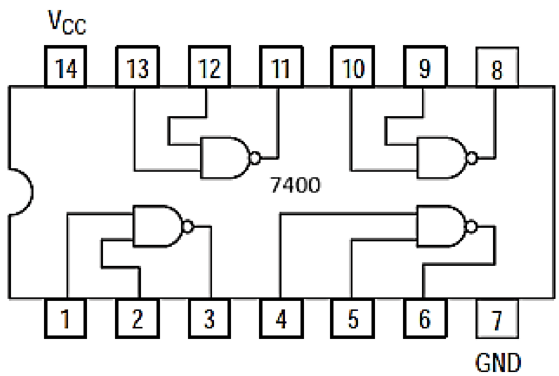
NAND gate is universal gate. It can perform all the basic logic function. NAND means NOT AND that is, AND output is NOTed.so NAND gate is combination of an AND gate and aNOT gate. The output is logic 0 level, only when each of its inputs assumes a logic 1 level. Forany other combination of inputs, the output is logic 1 level. NAND gate is equivalent to a bubbled OR gate.

The logic symbol & truth table of two input NAND gate are shown in figure.



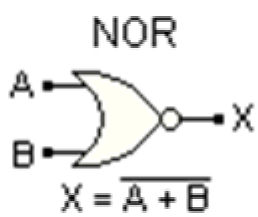
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

Pin diagram for IC**74LS00**:

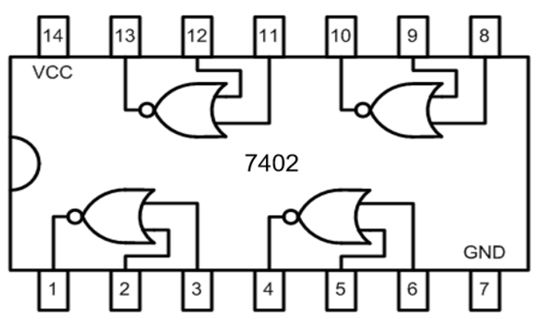


NOR GATE

NOR gate is universal gate. It can perform all the basic logic function. NOR means NOT OR that is, OR output is NOTed.so NOR gate is combination of an OR gate and a NOT gate. Theoutput is logic 1 level, only when each of its inputs assumes a logic 0 level. For any other combination of inputs, the output is logic 0 level. NOR gate is equivalent to a bubbled AND gate. The logic symbol & truth table of two inputs NOR gate are shown in figure



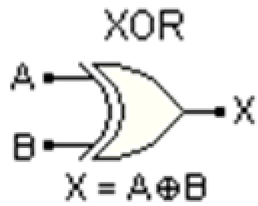
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0



Pin diagram for IC**74LS02**:

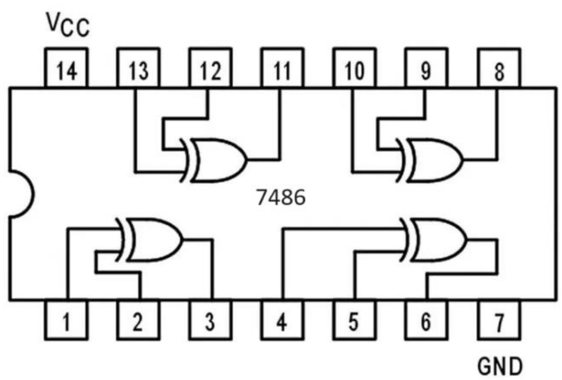
EXCLUSIVE-OR (X-OR) GATE

An X-OR gate is a two input, one output logic circuit, whose output assumes a logic 1 statewhen one and only one of its two inputs assumes a logic 1 state. Under the condition when both the inputs are same either 0 or 1, the output assumes a logic 0 state. Since an X-OR gate produces an output 1 only when the inputs are not equal, it is called as an anti-coincidence gate or inequality detector. The output of an X-OR gate is the modulo sum of its two inputs. The logicsymbol & truth table of two input X-OR gate are shown in figure.



A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

Pin diagram for IC**74LS86**:



RESULT: Truth Tables of various gates were verified.

PRECAUTIONS:

1. Connections should be right & tight.
2. Always take accurate reading.
3. Meters used should be without error.
4. Be alert while doing practical.
5. Never avoid short circuit & loose connection.
6. Never exceed the permissible value of current, Voltage and power of any instruments.
7. Check all the connections before switch on the power supply.

VIVA VOCE:

1. Which are the universal gates and why are they so called?
2. Which are basic logic gates?
3. Which gate gives compliment of input?
4. What is the output of NAND gate at high inputs?
5. What is the outputs of NOR gate at high inputs?